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Problem 1

a)

Considering the double-precision numbers in the IEEE-standard for binary floating-point arithmetic ($p = 53$, $E_{max} = 1023$, $E_{min} = -1022$), we want to find the:

1. Largest positive number
2. Smallest positive *normalized* number
3. Smallest positive number *overall*
4. Machine epsilon

The first step to obtain the above was to determine the value of f , or the fractional part of the numbers' representation. $f = p - 1$, so $f = 52$. This means that the normalized numbers all have fractional parts represented by 52 bits. The largest number would utilize E_{max} as the exponent and have 1's in each place in binary notation. That is, $(1.111 \dots 111)_2$, thus the largest value is $2^{1023} \cdot (1.111 \dots 111)_2$. Conversely, the smallest positive normalized number would utilize E_{min} as the exponent and have 0's in each binary location *aside* from the last one (so that the number is not 0). This gives $2^{-1022} \cdot (1.000 \dots 001)_2$. The smallest positive number overall is obtained similarly, the only difference is the leading value in binary: $2^{-1022} \cdot (0.000 \dots 001)_2$. Lastly, machine epsilon ϵ is defined as the smallest positive number such that $1 + \epsilon$ does not round to 1. The value is given by $2^{-(p-1)} \rightarrow 2^{-52}$, with two as the base because the system is binary. Base 10, trivially, would use 10 as the base for machine epsilon.

For legibility, the numbers were arranged in the order listed above:

1. $2^{1023} \cdot (1.111 \dots 111)_2 = 2^{1023} \cdot (2 - 2^{-52}) = 1.797693134 \cdot 10^{308}$
2. $2^{-1022} \cdot (1.000 \dots 001)_2 = 2^{-1022} \cdot (1 + 2^{-52}) = 2.225073858 \cdot 10^{-308}$
3. $2^{-1022} \cdot (0.000 \dots 001)_2 = 2^{-1022} \cdot 2^{-52} = 2^{-1074} = 4.940654584 \cdot 10^{-324}$
4. $2^{-52} = 2.220446049 \cdot 10^{-16}$

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b)

To confirm the results from part a), Octave was utilized

1 format hex	<code>ans = 3cb0000000000000</code>
2 eps	<code>ans = 0010000000000000</code>
3 realmin	<code>ans = 7fefffffffffffffff</code>
4 realmax	

In decimal format, the previous values are given by

```
1| format long
2| eps      ans = 2.220446049250313e-16
3| realmax  ans = 1.797693134862316e+308
4| realmin  ans = 2.225073858507201e-308
```

These results agree with those found in part a), where realmax is equivalent to the largest positive number, realmin is equivalent to the smallest positive normalized number, and eps is of course equivalent to epsilon.

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Problem 2

a)

We now consider a binary floating-point computer number system that uses a total of 8 bits: 1 for the sign, 3 for the characteristic field, and 4 for the fractional part. Using this information, the following can be determined:

p :

$$f = p - 1$$
$$p = f + 1 = 4 + 1 = 5 \text{ degrees of precision}$$

E_{min} and E_{max} :

The largest number of the system is given by $(111)_2 = 7$ based on the number of bits for the characteristic field. This suggests that e is bounded by 0 and 7, but in order to make the system as symmetric as possible, $|E_{min}|$ should be close to E_{max} . This gives two options:
– $-2 \leq e \leq 3$ or $-3 \leq e \leq 2$. Using the IEEE standard, E_{max} is slightly larger than $|E_{min}|$, so the former option should be selected. This gives $E_{min} = -2$ and $E_{max} = 3$

Bias:

The IEEE standard shows that the bias is equivalent to E_{max} for the “double” format, thus the bias is also equal to 3.

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b)

This number system would have 192 total normalized with 96 of them positive, 15 positive *denormalized* numbers, and up to 6 other numbers (+inf, -inf, +0, -0, two NaN's).

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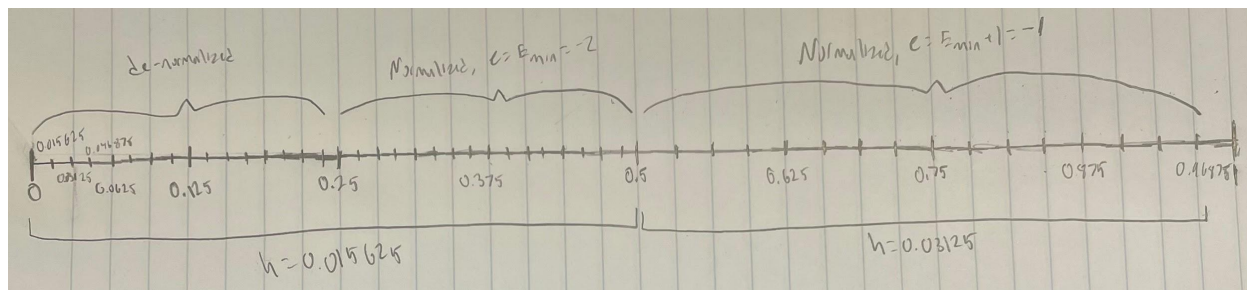
c)

Below are lists of all the positive numbers of the system, ordered from smallest to largest. The denormalized numbers are specified, along with the value of e . To save space and avoid redundancy, the binary notations were represented by a green line. This could be thought of as a column vector of sorts, and is explicitly written out in the green box in the image.

		$\Delta h = 0.015625$		$\Delta h = 0.03125$	
0		$2^{-2} \cdot (1.0000)_2 = 0.250000$		$2^{-1} = 0.50000$	
$2^{-2} \cdot (1.0001)_2 = 0.015625$		$2^{-2} \cdot (1.0001)_2 = 0.265625$		$2^{-1} = 0.53125$	
$2^{-2} \cdot (1.0010)_2 = 0.031250$		$2^{-2} \cdot (1.0010)_2 = 0.281250$		$2^{-1} = 0.56250$	
$2^{-2} \cdot (1.0011)_2 = 0.046875$		$2^{-2} \cdot (1.0011)_2 = 0.296875$		$2^{-1} = 0.593750$	
$2^{-2} \cdot (1.0100)_2 = 0.062500$		$2^{-2} \cdot (1.0100)_2 = 0.312500$		$2^{-1} = 0.62500$	
$2^{-2} \cdot (1.0101)_2 = 0.078125$		$2^{-2} \cdot (1.0101)_2 = 0.328125$		$2^{-1} = 0.65625$	
$2^{-2} \cdot (1.0110)_2 = 0.093750$		$2^{-2} \cdot (1.0110)_2 = 0.343750$		$2^{-1} = 0.68750$	
$2^{-2} \cdot (1.0111)_2 = 0.109375$		$2^{-2} \cdot (1.0111)_2 = 0.359375$		$2^{-1} = 0.71875$	
$2^{-2} \cdot (1.1000)_2 = 0.125000$		$2^{-2} \cdot (1.1000)_2 = 0.375000$		$2^{-1} = 0.75000$	
$2^{-2} \cdot (1.1001)_2 = 0.140625$		$2^{-2} \cdot (1.1001)_2 = 0.390625$		$2^{-1} = 0.78125$	
$2^{-2} \cdot (1.1010)_2 = 0.156250$		$2^{-2} \cdot (1.1010)_2 = 0.406250$		$2^{-1} = 0.81250$	
$2^{-2} \cdot (1.1011)_2 = 0.171875$		$2^{-2} \cdot (1.1011)_2 = 0.421875$		$2^{-1} = 0.84375$	
$2^{-2} \cdot (1.1100)_2 = 0.187500$		$2^{-2} \cdot (1.1100)_2 = 0.437500$		$2^{-1} = 0.87500$	
$2^{-2} \cdot (1.1101)_2 = 0.203125$		$2^{-2} \cdot (1.1101)_2 = 0.453125$		$2^{-1} = 0.906250$	
$2^{-2} \cdot (1.1110)_2 = 0.218750$		$2^{-2} \cdot (1.1110)_2 = 0.468750$		$2^{-1} = 0.93750$	
$2^{-2} \cdot (1.1111)_2 = 0.234375$		$2^{-2} \cdot (1.1111)_2 = 0.484375$		$2^{-1} = 0.96875$	
$\Delta h = 0.0625$		$h = 0.125$		$\Delta h = 0.25$	
$2^0 = 1.0000$		$2^1 = 2.000$		$2^2 = 4.00$	
$2^0 = 1.0625$		$2^1 = 2.125$		$2^2 = 4.25$	
$2^0 = 1.1250$		$2^1 = 2.250$		$2^2 = 4.50$	
$2^0 = 1.1875$		$2^1 = 2.375$		$2^2 = 4.75$	
$2^0 = 1.2500$		$2^1 = 2.500$		$2^2 = 5.00$	
$2^0 = 1.3125$		$2^1 = 2.625$		$2^2 = 5.25$	
$2^0 = 1.3750$		$2^1 = 2.750$		$2^2 = 5.50$	
$2^0 = 1.4375$		$2^1 = 2.875$		$2^2 = 5.75$	
$2^0 = 1.5000$		$2^1 = 3.000$		$2^2 = 6.00$	
$2^0 = 1.5625$		$2^1 = 3.125$		$2^2 = 6.25$	
$2^0 = 1.6250$		$2^1 = 3.250$		$2^2 = 6.50$	
$2^0 = 1.6875$		$2^1 = 3.375$		$2^2 = 6.75$	
$2^0 = 1.7500$		$2^1 = 3.500$		$2^2 = 7.00$	
$2^0 = 1.8125$		$2^1 = 3.625$		$2^2 = 7.25$	
$2^0 = 1.8750$		$2^1 = 3.750$		$2^2 = 7.50$	
$2^0 = 1.9375$		$2^1 = 3.875$		$2^2 = 7.75$	
$2^0 = 2.0000$		$2^1 = 4.000$		$2^2 = 8.00$	
$2^0 = 2.0625$		$2^1 = 4.125$		$2^2 = 8.25$	
$2^0 = 2.1250$		$2^1 = 4.250$		$2^2 = 8.50$	
$2^0 = 2.1875$		$2^1 = 4.375$		$2^2 = 8.75$	
$2^0 = 2.2500$		$2^1 = 4.500$		$2^2 = 9.00$	
$2^0 = 2.3125$		$2^1 = 4.625$		$2^2 = 9.25$	
$2^0 = 2.3750$		$2^1 = 4.750$		$2^2 = 9.50$	
$2^0 = 2.4375$		$2^1 = 4.875$		$2^2 = 9.75$	
$2^0 = 2.5000$		$2^1 = 5.000$		$2^2 = 10.00$	
$2^0 = 2.5625$		$2^1 = 5.125$		$2^2 = 10.25$	
$2^0 = 2.6250$		$2^1 = 5.250$		$2^2 = 10.50$	
$2^0 = 2.6875$		$2^1 = 5.375$		$2^2 = 10.75$	
$2^0 = 2.7500$		$2^1 = 5.500$		$2^2 = 11.00$	
$2^0 = 2.8125$		$2^1 = 5.625$		$2^2 = 11.25$	
$2^0 = 2.8750$		$2^1 = 5.750$		$2^2 = 11.50$	
$2^0 = 2.9375$		$2^1 = 5.875$		$2^2 = 11.75$	
$2^0 = 3.0000$		$2^1 = 6.000$		$2^2 = 12.00$	
$2^0 = 3.0625$		$2^1 = 6.125$		$2^2 = 12.25$	
$2^0 = 3.1250$		$2^1 = 6.250$		$2^2 = 12.50$	
$2^0 = 3.1875$		$2^1 = 6.375$		$2^2 = 12.75$	
$2^0 = 3.2500$		$2^1 = 6.500$		$2^2 = 13.00$	
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$2^0 = 3.6250$		$2^1 = 7.250$		$2^2 = 14.50$	
$2^0 = 3.6875$		$2^1 = 7.375$		$2^2 = 14.75$	
$2^0 = 3.7500$		$2^1 = 7.500$		$2^2 = 15.00$	
$2^0 = 3.8125$		$2^1 = 7.625$		$2^2 = 15.25$	
$2^0 = 3.8750$		$2^1 = 7.750$		$2^2 = 15.50$	
$2^0 = 3.9375$		$2^1 = 7.875$		$2^2 = 15.75$	
$2^0 = 4.0000$		$2^1 = 8.000$		$2^2 = 16.00$	
$2^0 = 4.0625$		$2^1 = 8.125$		$2^2 = 16.25$	
$2^0 = 4.1250$		$2^1 = 8.250$		$2^2 = 16.50$	
$2^0 = 4.1875$		$2^1 = 8.375$		$2^2 = 16.75$	
$2^0 = 4.2500$		$2^1 = 8.500$		$2^2 = 17.00$	
$2^0 = 4.3125$		$2^1 = 8.625$		$2^2 = 17.25$	
$2^0 = 4.3750$		$2^1 = 8.750$		$2^2 = 17.50$	
$2^0 = 4.4375$		$2^1 = 8.875$		$2^2 = 17.75$	
$2^0 = 4.5000$		$2^1 = 9.000$		$2^2 = 18.00$	
$2^0 = 4.5625$		$2^1 = 9.125$		$2^2 = 18.25$	
$2^0 = 4.6250$		$2^1 = 9.250$		$2^2 = 18.50$	
$2^0 = 4.6875$		$2^1 = 9.375$		$2^2 = 18.75$	
$2^0 = 4.7500$		$2^1 = 9.500$		$2^2 = 19.00$	
$2^0 = 4.8125$		$2^1 = 9.625$		$2^2 = 19.25$	
$2^0 = 4.8750$		$2^1 = 9.750$		$2^2 = 19.50$	
$2^0 = 4.9375$		$2^1 = 9.875$		$2^2 = 19.75$	
$2^0 = 5.0000$		$2^1 = 10.000$		$2^2 = 20.00$	
$2^0 = 5.0625$		$2^1 = 10.125$		$2^2 = 20.25$	
$2^0 = 5.1250$		$2^1 = 10.250$		$2^2 = 20.50$	
$2^0 = 5.1875$		$2^1 = 10.375$		$2^2 = 20.75$	
$2^0 = 5.2500$		$2^1 = 10.500$		$2^2 = 21.00$	
$2^0 = 5.3125$		$2^1 = 10.625$		$2^2 = 21.25$	
$2^0 = 5.3750$		$2^1 = 10.750$		$2^2 = 21.50$	
$2^0 = 5.4375$		$2^1 = 10.875$		$2^2 = 21.75$	
$2^0 = 5.5000$		$2^1 = 11.000$		$2^2 = 22.00$	
$2^0 = 5.5625$		$2^1 = 11.125$		$2^2 = 22.25$	
$2^0 = 5.6250$		$2^1 = 11.250$		$2^2 = 22.50$	
$2^0 = 5.6875$		$2^1 = 11.375$		$2^2 = 22.75$	
$2^0 = 5.7500$		$2^1 = 11.500$		$2^2 = 23.00$	
$2^0 = 5.8125$		$2^1 = 11.625$		$2^2 = 23.25$	
$2^0 = 5.8750$		$2^1 = 11.750$		$2^2 = 23.50$	
$2^0 = 5.9375$		$2^1 = 11.875$		$2^2 = 23.75$	
$2^0 = 6.0000$		$2^1 = 12.000$		$2^2 = 24.00$	
$2^0 = 6.0625$		$2^1 = 12.125$		$2^2 = 24.25$	
$2^0 = 6.1250$		$2^1 = 12.250$		$2^2 = 24.50$	
$2^0 = 6.1875$		$2^1 = 12.375$		$2^2 = 24.75$	
$2^0 = 6.2500$		$2^1 = 12.500$		$2^2 = 25.00$	
$2^0 = 6.3125$		$2^1 = 12.625$		$2^2 = 25.25$	
$2^0 = 6.3750$		$2^1 = 12.750$		$2^2 = 25.50$	
$2^0 = 6.4375$		$2^1 = 12.875$		$2^2 = 25.75$	
$2^0 = 6.5000$		$2^1 = 13.000$		$2^2 = 26.00$	
$2^0 = 6.5625$		$2^1 = 13.125$		$2^2 = 26.25$	
$2^0 = 6.6250$		$2^1 = 13.250$		$2^2 = 26.50$	
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$2^0 = 6.9375$		$2^1 = 13.875$		$2^2 = 27.75$	
$2^0 = 7.0000$		$2^1 = 14.000$		$2^2 = 28.00$	
$2^0 = 7.0625$		$2^1 = 14.125$		$2^2 = 28.25$	
$2^0 = 7.1250$		$2^1 = 14.250$		$2^2 = 28.50$	
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$2^0 = 7.2500$		$2^1 = 14.500$		$2^2 = 29.00$	
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$2^0 = 7.5000$		$2^1 = 15.000$		$2^2 = 30.00$	
$2^0 = 7.5625$		$2^1 = 15.125$		$2^2 = 30.25$	
$2^0 = 7.6250$		$2^1 = 15.250$		$2^2 = 30.50$	
$2^0 = 7.6875$		$2^1 = 15.375$		$2^2 = 30.75$	
$2^0 = 7.7500$		$2^1 = 15.500$		$2^2 = 31.00$	
$2^0 = 7.8125$		$2^1 = 15.625$		$2^2 = 31.25$	
$2^0 = 7.8750$		$2^1 = 15.750$		$2^2 = 31.50$	
$2^0 = 7.9375$		$2^1 = 15.875$		$2^2 = 31.75$	
$2^0 = 8.0000$		$2^1 = 16.000$		$2^2 = 32.00$	
$2^0 = 8.0625$		$2^1 = 16.125$		$2^2 = 32.25$	
$2^0 = 8.1250$		$2^1 = 16.250$		$2^2 = 32.50$	
$2^0 = 8.1875$		$2^1 = 16.375$		$2^2 = 32.75$	
$2^0 = 8.2500$		$2^1 = 16.500$		$2^2 = 33.00$	
$2^0 = 8.3125$		$2^1 = 16.625$		$2^2 = 33.25$	
$2^0 = 8.3750$		$2^1 = 16.750$		$2^2 = 33.50$	
$2^0 = 8.4375$		$2^1 = 16.875$		$2^2 = 33.75$	
$2^0 = 8.5000$		$2^1 = 17.000$		$2^2 = 34.00$	
$2^0 = 8.5625$		$2^1 = 17.125$		$2^2 = 34.25$	
$2^0 = 8.6250$		$2^1 = 17.250$		$2^2 = 34.50$	
$2^0 = 8.6875$		$2^1 = 17.375$		$2^2 = 34.75$	
$2^0 = 8.7500$		$2^1 = 17.500$		$2^2 = 35.00$	
$2^0 = 8.8125$		$2^1 = 17.625$		$2^2 = 35.25$	
$2^0 = 8.8750$		$2^1 = 17.750$		$2^2 = 35.50$	
$2^0 = 8.9375$		$2^1 = 17.875$		$2^2 = 35.75$	
$2^0 = 9.0000$		$2^1 = 18.000$		$2^2 = 36.00$	
$2^0 = 9.0625$		$2^1 = 18.125$		$2^2 = 36.25$	
$2^0 = 9.1250$		$2^1 = 18.250$		$2^2 = 36.50$	
$2^0 = 9.1875$		$2^1 = 18.375$		$2^2 = 36.75$	
$2^0 = 9.2500$		$2^1 = 18.500$		$2^2 = 37.00$	
$2^0 = 9.3125$		$2^1 = 18.625$		$2^2 = 37.25$	
$2^0 = 9.3750$		$2^1 = 18.750$		$2^2 = 37.50$	
$2^0 = 9.4375$		$2^1 = 18.875$		$2^2 = 37.75$	

d)

The following number line can be used to visualize the dispersion of the numbers in the system. Each vertical line represents a number in the system, and the figure itself only displays the numbers associated with E_{min} , $E_{min} + 1$, and the denormalized numbers.



Clearly, the denormalized numbers and the numbers associated with E_{min} are spaced more closely and are thus “less scaled”. Each of these numbers are separated from its neighbors by 0.015625. Notably, the gap between 0 (which does not fall under the denormalized category) and the first number is also equivalent to this. In other words, there are no jumps. However, larger jumps begin to take place for the normalized numbers associated with $E_{min} + 1$, where each successive number is separated by 0.03125. The scale has increased, specifically by a factor of 2, although the quantity of numbers associated with the respective e has not changed. This behavior could be further visualized with the inclusion of numbers associated with $E_{min} + 2$, where the gap between numbers doubles again to 0.0625. It can be said that the closer e grows to E_{max} , the larger the range of values covered by the numbers

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e)

With this system, it can be seen that the largest positive number is 15.5. The smallest positive normalized number is 0.25, and the smallest positive number overall is 0.015625. Machine epsilon is 2^{-7} , given by the same formula used in Problem 1a. For that matter, all of these results were obtained in a similar method to Problem 1a, where the largest positive number was given by $2^3(1.111)_2$, smallest positive normalized is $2^{-2} \cdot (1.001)_2$, and smallest overall is $2^{-2} \cdot (1.001)_2$.

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Problem 3

The objective of this problem is to develop an efficient algorithm to create a vector x with components $x_i = (i - 1)h$, $i = 1, 2, \dots, N$ with $h = \frac{1}{N-1}$ for some positive integer $N \geq 2$.

Octave code may be used to implement this algorithm. In addition to the creation of the vector, the code will return the last component of the vector, along with the difference between said component and its intended value of 1. The following code is in three parts, the driver code to call the functions, and the functions themselves defined in separate files. The code will be examined using $N = 101$. It will be clear that an accumulated round-off has occurred, hence the name of the first function.

Alex Caramanico May 11, 2023. hw9p3.m

```
1| format long
2| n = 101;
3| disp('Bad Algorithm')
4| badalg(n)
5| disp('Better Algorithm')
6| goodalg(n)
```

badalg.m

```
1| function [last, diff] = badalg(N);
2| h = 1 / (N-1);
3| x = zeros(1,N);
4| for i = 2:N
5|     x(i) = x(i-1)+h;
6| endfor
7| last = x(N)
8| diff = last - 1
9| end
```

goodalg.m

```
1| function [last, diff] = goodalg(N)
2| x = zeros(1,N);
3| for i = 1:N-1
4|     h = i/(N-1);
5|     x(i+1) = x(1) + h;
6| endfor
7| last = x(N)
8| diff = last - 1
9| end
```

```
Bad Algorithm
last = 1.0000000000000001
diff = 6.661338147750939e-16
ans = 1.0000000000000001
Better Algorithm
last = 1
diff = 0
ans = 1
```

We see that the first algorithm is imprecise. The last value was expected to be exactly equal to 1, and therefore the difference between said value and 1 was expected to be 0. The second algorithm is precise though, with the output exactly matching the expected values.

The first algorithm does not perform as well because of the accumulated roundoff. This occurs because the vector is being updated

according to its previous entry, so the amount of round off increases at each step of the for loop. The second algorithm, on the other hand, only uses the first entry of the vector which is set to be zero. The for loop then adds to the first number according to the iteration number, with larger jumps from 0 at later steps as i increments. This means that the better algorithm avoids accumulated roundoff because each entry in the vector is created independently, as opposed to the first algorithm where each entry was dependent on the previous entry.

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